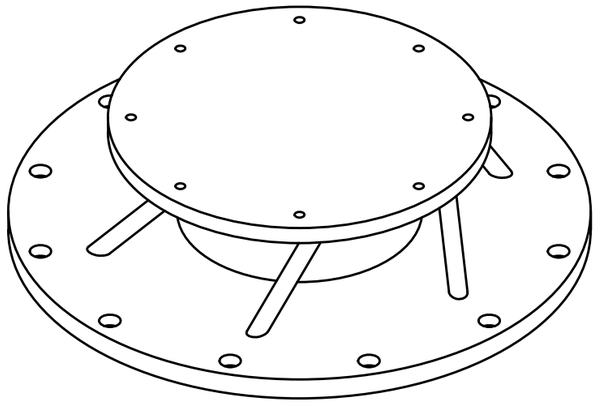
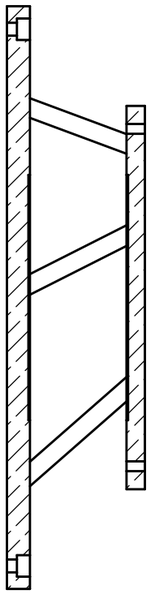
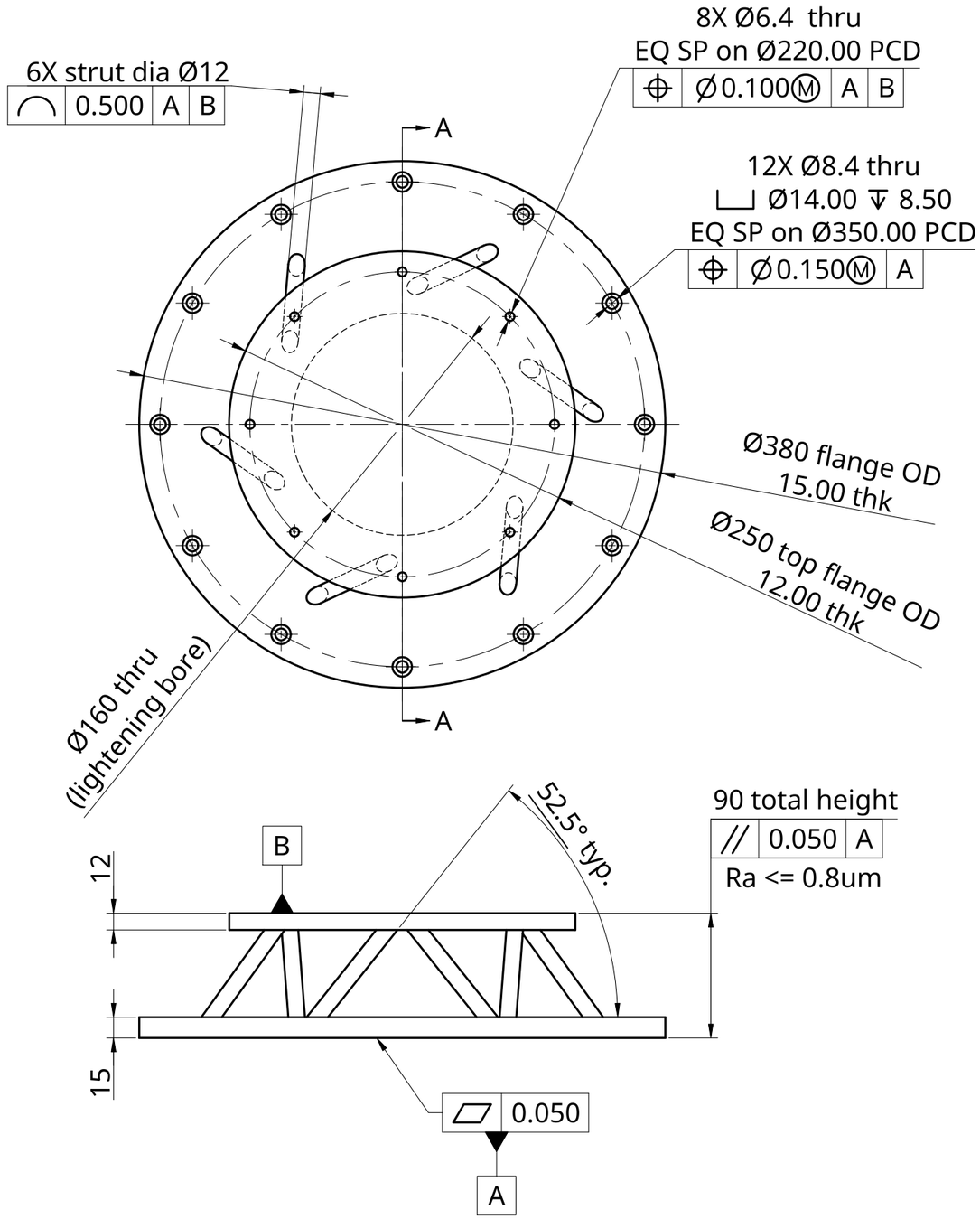


REFERENCE ONLY



A - A



NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W1-AERO-SAT-ADP-350_RevA.STEP) GOVERNS NOMINAL GEOMETRY.
   1.3 GENERAL LINEAR TOLERANCES: +/- 0.1 mm, ANGULAR: +/- 0.5 DEG (PER ISO 2768-m).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

2. MATERIAL & THERMAL POST-PROCESSING:
   2.1 PRIMARY MATERIAL: AEROSPACE GRADE Ti-6Al-4V (GRADE 5) PER AMS 4928 / ASTM F1472.
   2.2 VACUUM STRESS RELIEF: 650°C +/- 10°C IN VACUUM (< 1.0E-4 mbar) FOR 120 MINS, ARGON COOLED.
   2.3 HOT ISOSTATIC PRESSING (HIP): 920°C +/- 10°C AT 100 MPa FOR 120 MINS PER ASTM F3001.
   2.4 THEORETICAL CALCULATED NET MASS: 7.955 kg (NET SOLID VOLUME: 1.796E+06 mm^3).

3. MANUFACTURING & SURFACE CONTROLS:
   3.1 SELECTIVE LASER MELTING (SLM) INTEGRAL MONOLITHIC BUILD (NO SPLICES OR JOINTS).
   3.2 MOUNTING DATUM FACES (DATUM A & B) FINISHED VIA 1-SETUP 5-AXIS PRECISION MILLING.
   3.3 TOP AND BOTTOM MATING INTERFACES SURFACE ROUGHNESS Ra <= 0.8 MICRONS.
   3.4 ALL NON-MACHINED AS-BUILT TRUSS STRUTS Ra <= 6.3 MICRONS (MICRO-BLAST FINISHED).

4. NON-DESTRUCTIVE TESTING & AEROSPACE FLIGHT ASSURANCE:
   4.1 100% X-RAY RADIOGRAPHIC INSPECTION PER ASTM E1742 CLASS 1 TO VERIFY INTERNAL SOLIDITY.
   4.2 FLUORESCENT PENETRANT INSPECTION (FPI) PER ASTM E1417 (TYPE 1, METHOD A/D, LEVEL 3).
   4.3 STATIC PROOF LOAD: WITHSTAND 15G AXIAL COMPRESSION WITHOUT PERMANENT PLASTIC SET.

5. CLEANING & FLIGHT-READY PACKAGING:
   5.1 PRECISION SOLVENT DEGREASED AND ULTRASONICALLY CLEANED PER NASA-STD-6016.
   5.2 WRAPPED IN NON-SHEDDING ANTISTATIC POLYETHYLENE AND ENCLOSED IN RIGID SHIPPING CRADLE.
   5.3 EACH ADAPTER PERMANENTLY SERIALIZED VIA LASER ENGRAVING ON BASE FLANGE OD RIM.

		NAME	SIGNATURE	DATE
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS	DRAWN			2026-10-15
ANGULAR = ± °	CHECKED			
SURFACE FINISH	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
THIRD ANGLE PROJECTION	MATERIAL: Ti-6Al-4V Grade 5 per AMS 4928 / ASTM F1472		FINISH	

TITLE: SPATIAL TRUSS SATELLITE INTERSTAGE ADAPTER

SIZE	DWG NO.	REV.
A3	W1-AERO-SAT-ADP-350_REVA	A

SCALE	WEIGHT	SHEET
NTS	7.955 kg	1 of 1